End-of-Year Test, Grade 1

This test is comprehensive, thus quite long. I do not recommend that you have your child/student do it in one sitting. Break it into parts and administer them either on consecutive days, or perhaps on morning/evening/morning. Use your judgment.

This is to be used as a diagnostic test. Thus, you may skip those areas and concepts that you already know for sure your student has mastered.

The test does not cover every single concept that is covered in *Math Mammoth Grade 1*, but all of the major concepts and ideas are tested here. This test is evaluating the child's ability in the following content areas:

- basic addition and subtraction facts within 0-10
- two-digit numbers
- adding and subtracting two-digit numbers
- basic word problems
- clock to the nearest half hour
- geometry, measuring, time, and fractions
- counting coins

Note 1: If the child cannot read, the teacher can read the questions.

Note 2: Problems #1 and #2 are done orally and timed. Let the student see the problems. Read each problem aloud, and wait a maximum of 5 seconds for an answer. Mark the problem as right or wrong according to the student's (oral) answer. Mark it wrong if there is no answer. Then you can move on to the next problem.

You do not have to mention to the student that the problems are timed or that he/she will have 5 seconds per answer, because the idea here is not to create extra pressure by the fact it is timed, but simply to check if the student has the facts memorized (quick recall). You can say for example (vary as needed):

"*I will ask you some addition and subtraction questions. Try to answer them as quickly as possible. In each question, I will only wait a little while for you to answer, and if you don't say anything, I will move on to the next problem. So just try your best to answer the questions as quickly as you can.*"

In order to continue with Math Mammoth Grade 2, I recommend that the child gain a minimum score of 80% on this test, and that the teacher or parent review with him any content areas that are found weak. Children scoring between 70 and 80% may also continue with grade 2, depending on the types of errors (careless errors or not remembering something, vs. lack of understanding). Again, use your judgment.

My suggestion for grading is below. The total is 128 points. A score of 102 points is 80%. A score of 90 points is 70%.

Question	Max. points	Student score
Basic Addition and Subtraction Facts within 0-10		
1	8 points	
2	8 points	
3	4 points	
4	6 points	
5	4 points	
	subtotal	/ 30
Place Value and Two-Digit Numbers		
6	1 point	
7	6 points	
8	6 points	
9	4 points	
	subtotal	/ 17
Adding and Subtracting Two-Digit Numbers		
10	6 points	
11	6 points	
12	2 points	
13	4 points	
14	4 points	
	subtotal	/ 22
Basic Word Problems		
15	2 points	
16	1 point	
17	2 points	
18	2 points	
19	2 points	
20	2 points	
21	4 points	
	subtotal	/ 15

Question	Max. points	Student score
Clock		
22	3 points	
23	3 points	
24	3 points	
	subtotal	/ 9
Measuring		
25	3 points	
26	4 points	
27	3 points	
28	1 point	
	subtotal	/ 11
Geometry and Fractions		
29	2 points	
30	4 points	
31	2 points	
32	2 points	
33	3 points	
	subtotal	/ 13
Graphs		
34	3 points	
35	3 points	
	subtotal	/ 6
Money		
36	3 points	
37	2 points	
	subtotal	/ 5
TOTAL		**/ 128**

End-of-Year Test

Basic Addition and Subtraction Facts within 0-10

In problems 1 and 2, your teacher will read you some addition and subtraction questions. Try to answer them as quickly as possible. In each question, your teacher will only wait a little while for you to answer, and if you don’t say anything, your teacher will move on to the next problem. So, just try your best to answer the questions as quickly as you can.

1. Add.

a.	**b.**	**c.**	**d.**
2 + 3 = ______	7 + 3 = ______	6 + 2 = ______	5 + 5 = ______
4 + 4 = ______	5 + 4 = ______	4 + 6 = ______	2 + 4 = ______
1 + 6 = ______	3 + 6 = ______	2 + 5 = ______	9 + 1 = ______
2 + 7 = ______	1 + 7 = ______	6 + 2 = ______	5 + 3 = ______

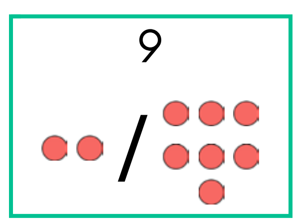
2. Subtract.

a.	**b.**	**c.**	**d.**
8 – 3 = ______	5 – 3 = ______	7 – 3 = ______	10 – 3 = ______
6 – 4 = ______	7 – 4 = ______	9 – 4 = ______	5 – 4 = ______
10 – 6 = ______	9 – 6 = ______	4 – 3 = ______	8 – 6 = ______
8 – 7 = ______	6 – 3 = ______	10 – 7 = ______	9 – 7 = ______

3. Write a fact family to match the picture.

_____ + _____ = _____   _____ + _____ = _____

_____ – _____ = _____   _____ – _____ = _____



4. Fill in the missing numbers.

a. 2 + ______ = 7	**b.** ______ + 1 = 8	**c.** 6 – ______ = 4
d. 3 + ______ = 8	**e.** ______ – 2 = 5	**f.** ______ – 3 = 4

5. Are these equations correct? Cross out the ones that are not.

a. 3 + 5 = 9 **b.** 10 = 8 + 3 **c.** 2 = 8 – 6 **d.** 7 – 5 = 3

Place Value and Two-Digit Numbers

6. Write the number that has 7 ones and 3 tens.

7. Fill in the missing numbers.

a. 20 + 7 = ________ 5 + 90 = ________	**b.** 56 = 6 + ________ 39 = 30 + _______	**c.** 100 + 2 = ________ 90 + 10 = ________

8. Add and subtract.

a. 24 + 10 = ________ 68 + 10 = ________	**b.** 45 – 10 = ________ 92 – 10 = ________	**c.** 50 – 30 = ________ 40 + 50 = ________

9. Compare the numbers. Write < or > in the box.

a. 48 ☐ 84 **b.** 70 ☐ 17 **c.** 59 ☐ 55 **d.** 3 + 40 ☐ 34

Adding and Subtracting Two-Digit Numbers

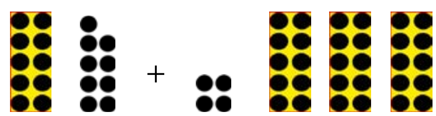
10. Add.

a. 84 + 4 = ________ 41 + 4 = ________	**b.** 6 + 70 = ________ 16 + 2 = ________	**c.** 74 + 5 = ________ 6 + 53 = ________

11. Subtract.

a. 80 – 30 = ________ 17 – 3 = ________	**b.** 55 – 3 = _______ 100 – 4 = _______	**c.** 29 – 3 = _______ 50 – 2 = _______

12. Write an addition to match the picture.

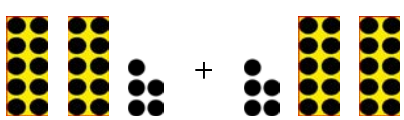


______ + _____ = _____

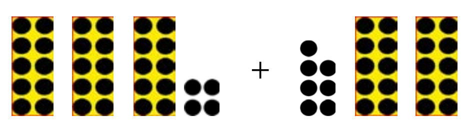
13. Add and subtract, writing the numbers in the grid.

a. 14 + 35 **b.** 59 – 34 **c.** 4 + 52 **d.** 96 – 6

14. Add. Use the images to help you.



a. 25 + 25 = _______



b. 34 + 27 =_______

Basic Word Problems

15. Write a subtraction sentence that matches with the addition 6 + 8 = 14.

______ – ______ = ______

16. How many more is 70 than 50? _______ more

17. Henry has four more cars than Mark. Mark has six cars.
Draw Mark’s and Henry’s cars.

Mark:

Henry:

18. Ten children (both boys and girls) are playing in the yard. There are six boys.
How many girls are there?

19. Andy drew eight stars. Then he drew three more stars and yet later, five more stars.
How many stars did Andy draw?

20. Mom baked 24 cookies. Luna ate three of them.
How many cookies are left now?

21. Robert has 12 toy cars and Luis has 7.

a. How many cars do the boys have together?

b. How many more cars does Robert have than Luis?

Clock

22. Tell or write the time using *o'clock* or *half past*.

		
a. ______________________	**b.** ______________________	**c.** ______________________
______________________	______________________	______________________

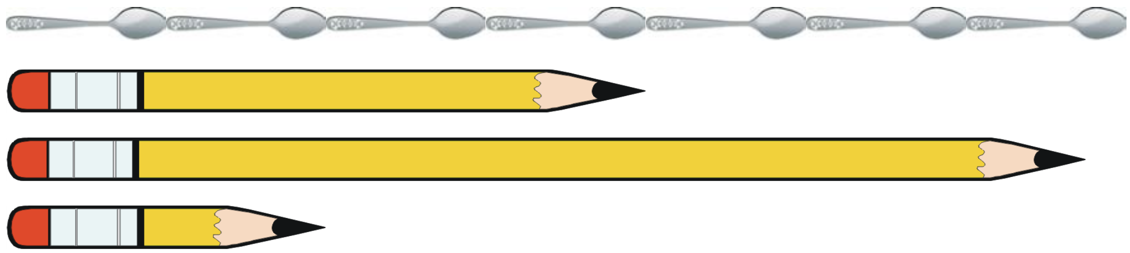
23. Match each digital clock with a clock below showing the same time.

24. Write the time a half-hour later from the given time. Use numbers.

Now it is:	**a.** 3:00	**b.** 12:00	**c.** 8:30
a half-hour later, it is:			

Measuring

25. How many little spoons long are these pencils?



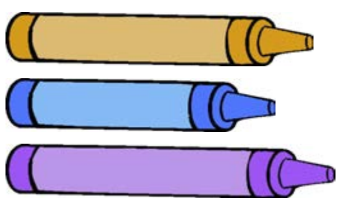
a.
b.
c.

a. ________ spoons long **b.** ________ spoons long **c.** ________ spoons long

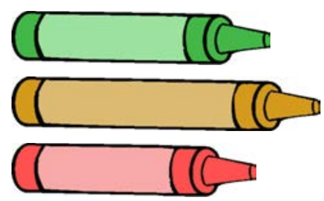
26. <u>Either</u> do the activity (i) or the written exercise (ii) below.

(i) Activity. Order three objects by length.
<u>You need</u>: Two sets of three objects of unequal length.
The student will order the three objects by length, from the shortest to the longest.

(ii) Order the crayons from the shortest to the longest, numbering them with 1, 2, and 3.



a.



b.

27. Measure these lines with an inch-ruler:

a.  ______ inches

b.  ______ inches

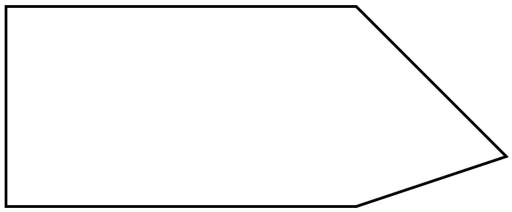
c.  ______ inches

28. Cut out (carefully!) the measuring stick from the bottom of this page and use it to compare the lengths of these things. Is the spoon or the screwdriver longer?

Cut out: 

Geometry and Fractions

29. Draw *one* line in each shape so that you get new shapes as indicated.



a. a rectangle and a triangle

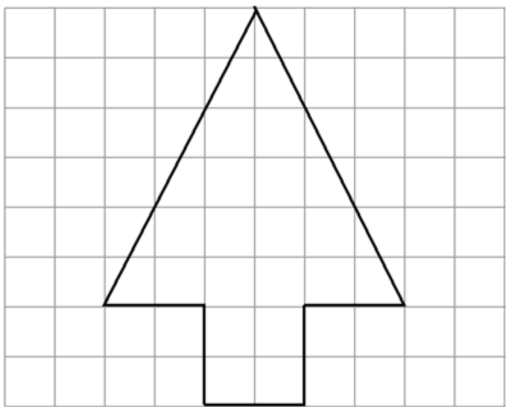
b. 2 four-sided shapes

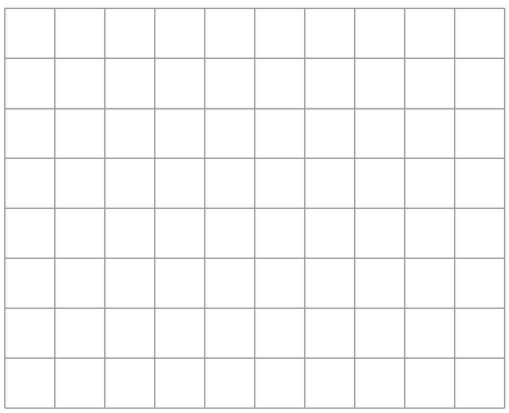
30. Draw (freehand, sketching) a figure as indicated.
(Your figure does not have to be accurate but just a sketch.)

a.
- The shape is a triangle.
- Each side is the same length.

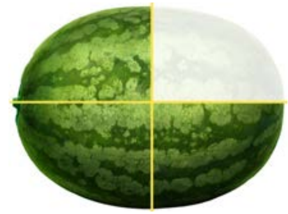
b.
- The shape has four sides.
- Each corner is a "square" corner (a right angle).

31. Copy the shape in the grid on the right.

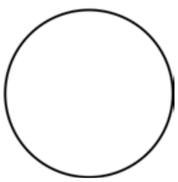
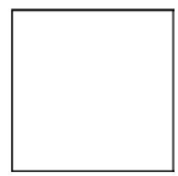




32. Tell or write what part (what fraction) of a fruit you see (e.g. two halves of a pear).

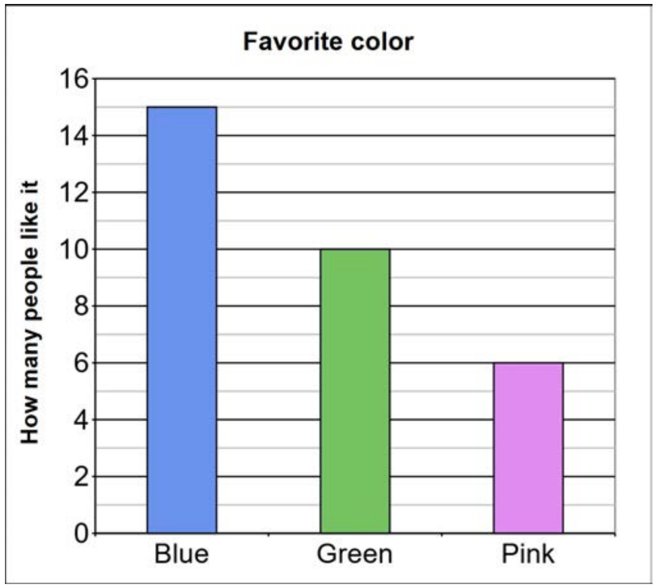
a.  ____________________ ____________________ of an apple	**b.**  ____________________ ____________________ of a watermelon

33. First divide the shape into equal shares, then color.

a. Color 2 halves. 	**b.** Color 1 quarter. 	**c.** Color 3 fourths. 

Graphs

34. The graph shows the favorite color of some people. Answer the questions.

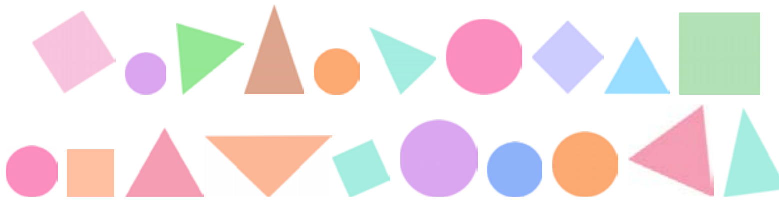


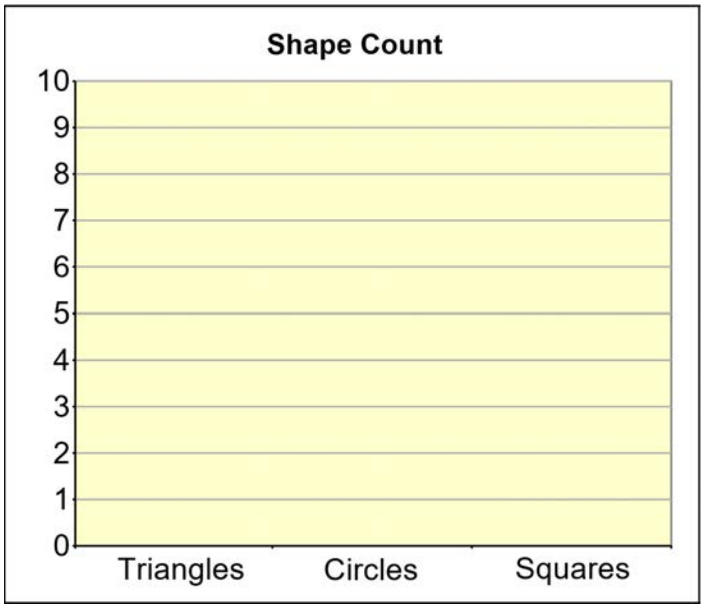
a. How many more people like green than like pink?

b. How many more people like blue than like green?

c. How many people like blue, green, or pink in total?

35. Draw a bar graph to show the count of each kind of shape (circles, triangles, squares).





Money

36. How much money? Write the amount in cents.

a.

_________ ¢

b.

_________ ¢

c.

_________ ¢

37. Solve.

You have:

You bought an apple for 35¢.

How much money do you have left? _________ ¢